BPC Response to NSF/OSTP RFI:
Development of an Artificial Intelligence (AI) Action Plan

TO: [redacted]
Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO), National Science Foundation (NSF)
2415 Eisenhower Avenue
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SUBJECT: "Development of an AI Action Plan"

FROM: Bipartisan Policy Center

The Bipartisan Policy Center appreciates the opportunity to offer insights in response to the National Science Foundation's (NSF) request for information on the Development of an Artificial Intelligence (AI) Action Plan. BPC brings together diverse perspectives to craft data-driven, pragmatic policy solutions.

For more than 20 years, BPC and Bipartisan Policy Center Action have been consistently achieving results on complex issues, demonstrating that bipartisan cooperation is not only possible but essential for lasting solutions. This letter reflects BPC's broad policy expertise on AI-related issues such as energy resiliency, innovation, national competitiveness, and workforce development.

These recommendations emphasize BPC's commitment to advancing AI innovation that strengthens U.S. global leadership, supports critical energy and infrastructure needs, safeguards privacy, and enhances security and economic competitiveness. We look forward to working with the administration to develop and implement a forward-looking AI Action Plan that meets the demands and opportunities of this rapidly evolving landscape.

I. Data Centers, Energy Consumption and Efficiency, Hardware and Chips

BPC's AI and Energy Project is actively exploring bipartisan policy opportunities at the intersection of AI, energy, and infrastructure. We are committed to forward-looking solutions that meet AI's growing electricity demands and accelerate AI infrastructure development in the United States.

In February 2025, BPC published *Electricity Demand Growth and Data Centers: A Guide for the Perplexed*, which examines the accuracy of data center electricity demand projections and their implications for policymakers. The report determines that data centers will likely account for up to 25% of expected electricity load growth nationally through 2030, with the remaining 75% being driven by onshoring of manufacturing and increasing electrification of transportation, heating, and industrial processes. It is also important to address future uncertainties from advancements or changes in computing efficiency, manufacturing production capacity, and demand for AI products and services.

BPC recommends the following actions for addressing the increasing needs of AI, energy efficiency, and buildout of critical infrastructure:

- ***Advance efficiencies in AI chip technology and data centers.*** Direct the Department of Energy (DOE) and other federal agencies to invest in advancements that improve the energy efficiency of data centers and next-generation chips. These innovations will support the expansion of critical infrastructure in an increasingly digitized economy reliant on new data centers and AI-driven technologies.

- ***Promote low-carbon power procurement for data centers.*** Foster DOE partnerships with data center operators to explore advanced nuclear, geothermal, carbon capture, and other clean firm power sources. Leveraging low-carbon solutions will help ensure a resilient power supply strategy that meets growing national energy demands, supports continuing AI advancements, and upholds U.S. global security and competition priorities.

- ***Incentivize utility and data center expansion.*** Establish grants and targeted tax credits to accelerate investment in generation capacity, transmission upgrades, and cybersecurity enhancements. A federally supported model that incentivizes states to accommodate new data center load will help keep the U.S. energy system resilient, affordable, and adaptable to increasing electricity demands. Reducing risks and supporting the development of new energy solutions can further strengthen these efforts.

- ***Develop standardized metrics for data centers***. Task the National Institute of Standards and Technology (NIST), in collaboration with DOE and industry stakeholders, with developing clear, standardized frameworks for measuring data center computing efficiency and power usage. Current measuring methods are imperfect, and no standard metric currently exists. Metrics for resource usage and efficiency would help policymakers and industry stakeholders better manage energy resources used by rapidly expanding AI workloads. A clear, standardized framework, combined with voluntary reporting, would also improve future energy demand forecasts; enable long-term planning benefits for industry, utilities, and the public; and enhance U.S. global competitiveness.

- ***Streamline key infrastructure buildout support for AI Energy Zones.*** Develop a strategy to streamline approvals in federally designated areas to enable rapid buildout of data centers powered by clean firm energy, paired with incentives that meet local needs. BPC is currently leading workshops with key leaders across industry, utilities, non-profits, and academia to identify and propose solutions for advancing the buildout of data centers and infrastructure.

- ***Prioritize affordability and reliability in AI energy infrastructure development.*** Establish critical guardrails that consider cost impacts and affordability for all customers (residential, customer, industrial). By doing so, dual objectives of supporting digital infrastructure and reducing energy bills for U.S. households and businesses can be achieved.

II. Research and Development, Innovation, and Competition

DOE has long demonstrated the unique expertise and capabilities necessary to study, use, and develop innovative AI applications. BPC recognizes, identifies, and encourages DOE's continued R&D investments and contributions to a robust national AI strategy.

Building on DOE's long history of pioneering investments in AI and BPC's September 2024 report, *Accelerating AI Sustainability and Innovation at the Department of Energy*, BPC recommends the following actions to ensure the strength of U.S. R&D leadership:

- ***Centralize DOE AI strategy and execution.*** Establish a central DOE office to coordinate and execute a DOE AI oversight strategy across the department and its 17 national labs. In President Donald Trump's first term, DOE's Artificial Intelligence and Technology Office (AITO) served an essential role in coordinating activities while DOE's Office of Critical and Emerging Technologies (CET) accelerated R&D, strategy development, and AI adoption. Preserving and expanding their functions and strategic oversight will enhance efficiency, avoid redundancy, and solidify U.S. global leadership in AI.

- ***Advance AI applications under DOE's broad mission space.*** Leverage DOE's expertise for solving big problems at scale to advance AI applications covering science, energy, and security. The department's ability to organize large, interdisciplinary projects to advance public-sector missions can utilize AI to speed up regulatory processes in energy, accelerate clean technology development, and enhance national security.

- ***Coordinate federal AI research agendas across DOE, NSF, and NIST***. By uniting efforts on fundamental computing, technology solutions, and scientific modeling, the federal government can reduce research duplication and drive quicker breakthroughs. Collaborations between the National Science Foundation (NSF), DOE, and NIST promote more efficient use of resources, speeds up real-world implementation of AI innovations, and solidifies U.S. leadership in high-impact technology sectors.

- ***Increase researcher access to DOE's AI facilities.*** Expand NSF's National Artificial Intelligence Research Resource (NAIRR) and similar initiatives to more researchers, allowing access to advanced computing facilities and testbeds. DOE's national labs are estimated to support 90% of NAIRR's computing capacity, which benefit small institutions and research teams in ways that democratize innovation, accelerate discoveries, and maximize the return on federal investments in AI infrastructure.

- ***Leverage DOE national laboratory partnerships.*** Strengthen DOE national lab partnerships by supporting collaboration pathways and leveraging existing national assets. At a time when preserving America's global leadership in science and technology is more difficult and consequential than ever, DOE's lab network stands out as a national treasure without parallel. The national labs have been pioneering computing and AI advances for decades, and they are vital to maintaining U.S. competitiveness in these critical industries.

- ***Establish public-private test beds for AI-driven electric grid solutions.*** Collaborative test environments, coordinated by DOE and industry partners, would de-risk emerging technologies before large-scale deployment. Such pilots allow both researchers and utilities to refine and validate AI innovations in realistic conditions, accelerating market adoption and fostering a more resilient and modernized grid.

- ***Expand AI R&D for real-time electric grid management***. Direct DOE to explore AI-driven tools that can enhance electric grid reliability and help operators anticipate and address adverse events such as extreme weather events and fluctuating demand. Potential use cases might include weather-informed load forecasting, predictive maintenance, and outage detection, leading to improvements in the reliability and resilience of critical energy systems.

- ***Support long-term AI R&D programs that fill private sector gaps.*** Direct DOE to prioritize and maintain strong partnerships and collaboration with industry in AI research, particularly in fields with limited private-sector investment or expertise. DOE's past investments in AI and supercomputing are successful thanks to sustained long-term support that has complemented private-sector resources and expertise. Maintaining DOE leadership and investment in this space fills critical innovation gaps, drives technological breakthroughs, and enhances U.S. global competitiveness in AI and related industries.

- ***Encourage open government data initiatives for AI research***. Direct federal agencies, such as the Government Services Administration (GSA) and NSF, to expand and enhance publicly available datasets, ensuring they are consistently formatted, documented, and accessible for AI developers. This would accelerate innovation by reducing the time researchers spend on data cleaning, ultimately promoting broader participation in AI-driven problem-solving. Additionally, DOE should ensure that data produced by its scientific user facilities are AI-ready and should adopt clear standards on data quality to more easily enable AI applications.

II. National Security, Cybersecurity, and Data Privacy

- ***Deploy AI for critical infrastructure security and threat detection***. Instruct DOE and the Department of Homeland Security's Cybersecurity and Infrastructure Security Agency to develop AI-based advanced anomaly detection tools that can help identify cyber or physical threats early in energy grid, pipeline, and industrial facility operations. This proactive defense significantly reduces disruptions to essential services, safeguards economic assets, and bolsters national security.

- ***Enforce robust data safeguards in the AI lifecycle.*** Integrate federal agency privacy-focused practices and secure data-sharing frameworks at each stage of AI development. By prioritizing responsible data stewardship and cybersecurity, the federal government reduces the risk of breaches, fosters public confidence, and creates a solid foundation for ongoing AI-driven improvements.

- ***Develop a federal privacy framework tailored to AI systems.*** Work with Congress and key agencies to propose a federal data privacy bill that accounts for the unique demands of AI, such as large-scale data collection and model training. Clarifying data protection requirements would reduce legal uncertainties, improve public trust, and allow responsible innovation.

III. Technical and Safety Standards, Procurement

- ***Empower NIST to develop voluntary AI standards and certification programs.*** Provide dedicated funding and authority for NIST to develop voluntary technical standards, testing protocols, and certification systems for AI systems across various industries. This would increase trust in AI products, streamline regulatory compliance, and bolster U.S. leadership in defining global best practices.

- ***Standardize federal AI procurement processes and guidelines***. Direct the Office of Management and Budget and GSA to establish consistent procurement rules for federal AI tools, ensuring transparent evaluation criteria, performance benchmarks, and vendor accountability. These standards will help improve cost-efficiency, reduce administrative hurdles, and accelerate modernization efforts across federal programs.

- ***Institute clear documentation and transparency requirements for government AI.*** Require agencies to document core design choices, data inputs, and performance metrics for AI used in public services, especially in high-impact applications. Transparency builds trust, mitigates risk, and advances tools that serve the public good.

IV. Education and the Workforce

The adoption of AI can drive innovation and economic growth. However, public opinion on AI is often fraught with misconceptions, debates, and concerns. Misinformation and mistrust remain significant barriers to adoption. When communities lack clear and accurate information, AI-enabled products and services may struggle to gain public support, limiting their potential impact.

Building an AI-literate population is essential for national competitiveness, public confidence, and workforce preparedness in an increasingly digital world. AI-literate individuals make more informed technology decisions, understanding when AI meets their needs and how to maximize its benefits while minimizing risks. To support this effort, BPC launched its AI 101 initiative for Congress in 2024, providing essential AI fundamentals, workshops, and resources to hundreds of Capitol Hill staffers. By equipping policymakers with the knowledge and resources they need, BPC helps advance forward-looking AI policies and opportunities.

To unlock AI's full potential, develop informed consumers, and build a skilled U.S. workforce capable of competing globally, BPC recommends the following actions:

- ***Expand upskilling programs for the federal workforce.*** Offer civil servants micro-credentials, short courses, and rotational assignments to rapidly boost AI competency. A more technologically adept workforce will accelerate AI adoption in government services, reduce reliance on external contractors, and ensure cohesive policy implementation.

- ***Establish training and grant programs for state and local readiness.*** Equip states and municipalities with the knowledge, tools, and funding to integrate AI responsibly into public services such as emergency response, transportation management, and workforce development. Federal support can bridge resource gaps, ensure consistent service quality nationwide, and help local governments keep pace with rapid AI advancements.

- ***Promote AI modules within core curricula at colleges and universities***. Encourage federal seed grants to develop, pilot, and scale AI learning modules across disciplines such as engineering, health sciences, and public policy. Integrating AI fundamentals into standard degree programs expands the pipeline of job-ready graduates and fosters cross-disciplinary innovation. Evaluating pilot modules will ensure they increase the pipeline and equip graduates with AI-relevant skills, improving training practice and cost effectiveness.

- ***Support existing DOE programs that are training the national AI workforce.*** Federal investment for AI education and workforce development is essential to harness AI's full potential and address its complex challenges. DOE programs such as Oak Ridge National Laboratory's AI Summer Institute for undergraduates and DOE's Computational Graduate Fellowship prepare individuals for productive and innovative careers in critical sectors like health care, manufacturing, and defense.

- ***Attract and retain foreign-born AI talent.*** Direct the Department of Labor to update its Schedule A Shortage Occupation list to expedite green card processing for AI-related occupations facing domestic labor shortages. A strong pipeline of foreign-born AI talent is critical for maintaining America's position as a global AI leader.

CONCLUSION

BPC stands ready to collaborate with OSTP and other federal partners in shaping a comprehensive AI Action Plan that promotes technological innovation, strengthens U.S. infrastructure resilience, ensures public trust, and supports a skilled workforce. We appreciate your consideration of these recommendations and look forward to continued engagement on these critical issues. If you have any questions or need further information, please contact our team.

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Sincerely,

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